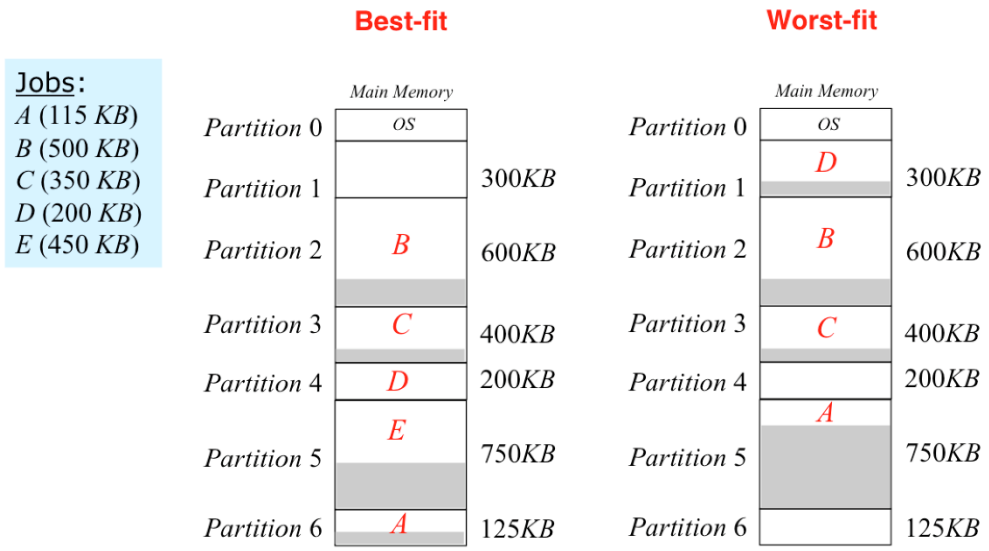
INSTRUCTION: Please answer all questions in the answer booklet. Show all your workings.

1. **[20 MARKS]** There are 5 processes (in order) of *A* (115 KB), *B* (500 KB), *C* (350 KB), *D* (200 KB), and *E* (450 KB) need to be placed in the memory.

   a) Suppose a fixed-partition is used with six memory partitions of 300 KB, 600 KB, 400 KB, 200 KB, 750 KB, and 125 KB (in order).

      (i) How would the best-fit and worst-fit algorithms place the processes? Draw the memory stack and label the partition size clearly. [10m]

Answer : Fixed-partition: Exercises 8.3 (page: 384)



      (ii) In terms of process placement in the memory, which algorithm faced a problem? What is the problem and state the cause of the problem. [4m]

Worst-fit.

Problem: Process E cannot be placed in the memory for Worst-fit

Cause: since there is no enough available partition.

   b) **[10 MARKS]** Suppose a variable or dynamic partition is used with the same 5 processes in Question 1. The following events happened:

t_1: process A and D left.

t_2: process F (50K) enters

t_3: process B left.

t_4: process G (550K) enters.

Draw the memory stack at t_2, t_3 and t_4 and label the sizes clearly for all processes. Use first-fit algorithm and relocatable dynamic partitions of merging adjacent holes to place the process in the memory. [6m]

F	50
	65
B	500
C	350
	200
E	450

T2

F	50
	565
C	350
	200
E	450

T3

F	50
G	550
	15
C	350
	200
E	450

T4

2. **[10 MARKS]** Answer the following questions.

a) Consider a logical address space of 2^8 pages with a 4 *KB* page size, mapped onto a physical memory of 4096 frames. Its page table is given in Table 2.

Table 2

0	50
1	30
2	400
3	100

(i) What is the size of logical and physical addresses (in 2^n bytes)? [4m]

page size → 4 KB = 2^{12}

- [2m] Logical address = totalpage x page size = 2^8 x 2^{12} = 2^{20} bytes
- [2m] Physical address = totalframe x frame size = (4096 pages) x (frame size = page size) =2^{12} x 2^{12} = 2^{24} bytes

(ii) Calculate the physical address if the logical address is 3 : 400. [3m]

3 → 100. Frame size = page size

Physical address = (frame# x frame size) + offset

= (100 x 2^{12}) + 400

= (100 x 4096) + 400

= 4096000 + 400

= 4096400

(iii) What is the page number and offset for the logical address of 51152? [3m]

51152/4096 = 12.48828125

Page Number: 12 – 1 = 11

Offset = (12.48828125 – 12) x 4096 = 2000 – 1 = 1999

3. **[12 MARKS]** Define the following terms.

   a) Demand paging [3m]

   Load pages only needed in virtual memory

   b) Lazy swapper

   Used in demand paging. Swaps a page into memory only when needed.

   c) Belady's anamoly [3m]

   Page fault rate may increase as the number of allocated frames increases, even though it looks like it should be the opposite effect.

   d) Thrashing in virtual memory [3m]

   High paging activity. Spending more time paging than executing.

4. **[13 MARKS]** Consider the following page reference string: **7, 2, 3, 1, 2, 5, 3, 4, 6, 7, 1, 3.** Assuming demand paging with four frames, how many page faults would occur for the following replacement algorithms? Please circle which references cause page faults.

   a) Optimal (OPT) replacement [6.5m]

7	2	3	1	2	5	3	4	6	7	1	3
7	7	7	7		7		7	7			
	2	2	2		5		4	6			
		3	3		3		3	3			
			1		1		1	1			

7 page fault

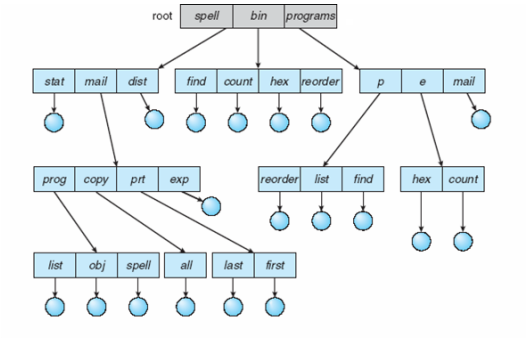
   b) LRU replacement [6.5m]

7	2	3	1	2	5	3	4	6	7	1	3
7	7	7	7		5		5	5	7	7	7
	2	2	2		2		2	6	6	6	6
		3	3		3		3	3	3	1	1
			1		1		4	4	4	4	3

10 page fault

5. **[10 MARKS]** Draw and discuss the advantages of:

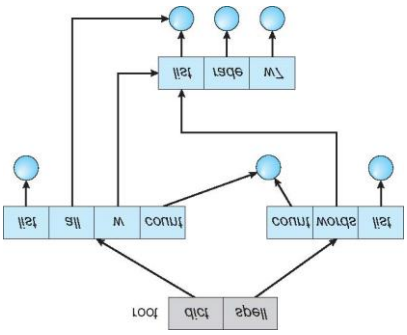
   a) Tree-structured directories [5m]



Text book 10.3.5, slide 13.21- 13.24

- Extention of two-level directory, tree can have arbitrary height
- Separate directory for each user – they can create their own subdirectory and organize their file accordingly.
- The tree has a root directory and every file has unique path name,
- Can have the same file name for different user
- Efficient searching
- Grouping capability

b) Acyclic-graph directories [5m]



Text book 10.3.6, slide 13.25

- Files/subdirectories have two different names (aliasing)
    - Only one actual file exists, so any changes made by one person are immediately visible to the other.
- How do we implement shared files and subdirectories?
    - Can duplicate the *inode* information about the shared file/subdirectory in each of the subdirectories that "point' to the shared structure.
        - If some information changes about the file it had to be updated in several places.
    - Have a new directory entry type:
        - **Link** – another name (pointer) to an existing file or subdirectory.

6. [**10 MARKS]** Answer the following questions:

a) In the Unix system, directory listing is shown as below:

```
-rw-r--r--      1 pbg staff      9423    Apr 19 2018   program
drwx------      3 pbg staff      1024    Aug 19 10:31  mail/
```

(i) Name and define the directory protection. [3m]

Directory name mail/
owner only allows to read write execute

(ii) Name and define the file protection. [3m]

Program owner can read and write ,
group can read only,
universe can read only

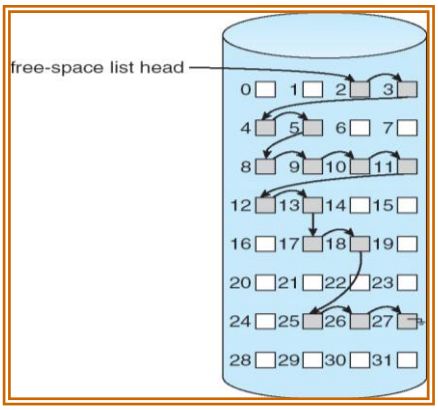
(iii) How to let all the users use the file without restriction? [3m]

chmod 777 program

7. **[10 MARKS]** Describe and show the example of free space management using:

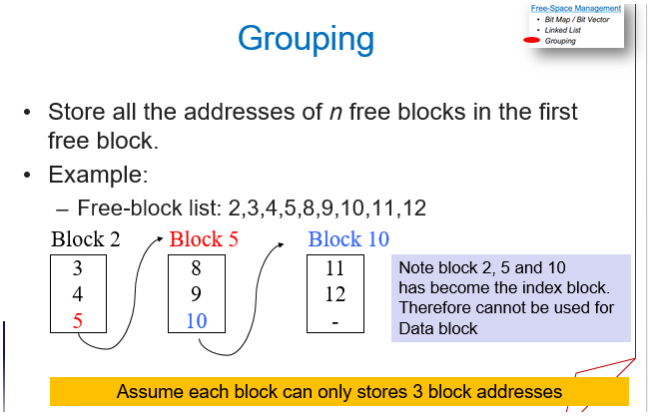
a) Linked list [5m]

To link together all the free disk block. Keeping a pointer to the first free block in a special location on the disk and caching it in memory.



b) Grouping [5m]

The first n-1 of these blocks are actually free. The last block contains the addresses of another n free blocks and so an. The addresses of a large number of free blocks can now be found quickly, unlike the linked-list

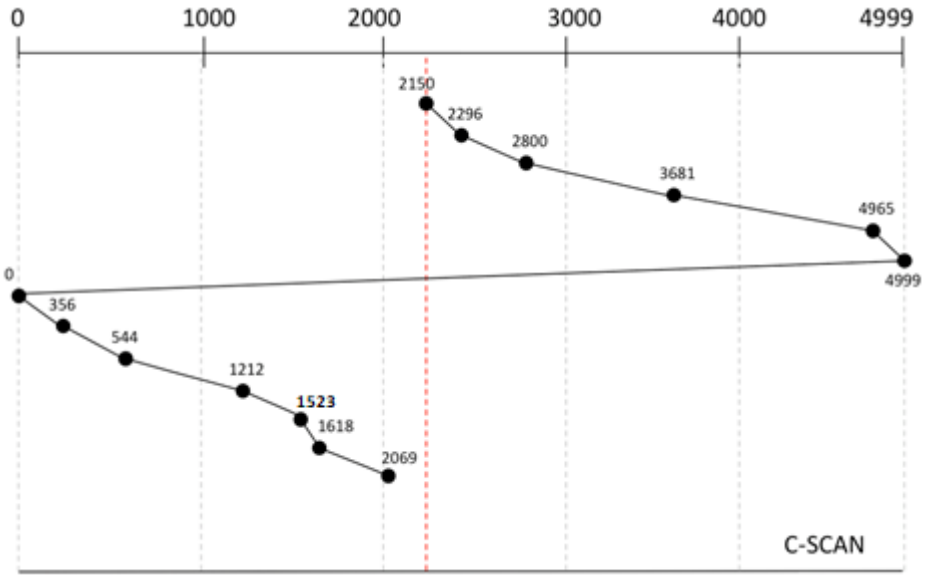


8. **[15 MARKS]** Consider the following information:

- A disk drive has 5000 cylinders, numbered 0 – 4999.
- Using C-SCAN disk-scheduling algorithm to satisfy all the cylinder requests.
- The drive is currently serving a request at cylinder 2150.
- The queue of pending requests is: **2069, 1212, 2296, 2800, 544, 1618, 356, 1523, 4965, 3681**

Starting from the current head position, answer the following question.

a) Draw the whole disk arm movement to satisfy all the pending requests. [5m]



b) What is the total seek distance (in cylinder) that the disk arm had moved? [2m]

4999 - 2150 = 2849

4999 - 0 = 4999

2069 - 0 = 2069

Total distance = 2849 + 4999 + 2069 = 9917 cylinders

c) Now, if C-LOOK algorithm is used to satisfy all the cylinder pending requests, compare the speed of both disk-scheduling algorithms based on the distance (in cylinder) that the disk arm had moved. [3m]

- C-LOOK uses total distance 780 less than S-SCAN
  - (4999-4965)x2 = 34 x 2 = 68
  - (356 x 2) = 712
  - total less = 68 + 712 = 780
- therefore, C-LOOK faster than S-SCAN